

LiDAR-Based Point Cloud Analysis Methodology for Tunnel Structural Health Monitoring

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Abstract

Keywords: *LiDAR; point cloud processing; tunnel structural health monitoring; ICP registration; Frenet-Serret frame; digital twin; automated deformation analysis*

Railway tunnel infrastructure is subject to progressive structural deterioration — including cross-sectional deformation, eccentricity, and lining convergence — that accumulates over decades of operation. Conventional visual inspection methods rely on subjective qualitative judgment and provide only snapshot-in-time assessments, rendering them inadequate for the quantitative, time-series evaluation required by modern asset management frameworks.

This paper proposes a comprehensive five-layer automated methodology for LiDAR-based point cloud analysis applied to tunnel structural health monitoring (SHM). The proposed pipeline integrates: (1) high-density 3D point cloud acquisition using mobile and static LiDAR systems; (2) intelligent preprocessing incorporating statistical distance-based noise removal and voxel-grid spatial homogenization; (3) high-precision time-series registration using intensity-derivative ring segmentation combined with iterative closest point (ICP) optimization over a kd-tree spatial index; (4) Frenet-Serret orthogonal coordinate system construction for geometrically correct cross-section extraction in curved tunnel alignments; and (5) automatic extraction of structural deformation parameters including vault settlement (δ_v), horizontal convergence (δ_h), eccentricity (e), ellipticity (ϵ), and radial deformation distribution (δ_r), visualized through continuous 3D heatmaps.

The system architecture further encompasses IFC-compliant BIM geometric conversion for integration with 5D lifecycle management platforms, and a local large language model (LLM) assistant embedded within a digital twin dashboard to support natural-language safety assessment and maintenance decision-making. The proposed methodology is designed for validation on the Osong Test Line railway tunnel, with a performance target of sub-2 mm registration accuracy (RMSE) and automated detection sensitivity at the millimeter scale.

By replacing manual inspection workflows with a standardized, reproducible automated pipeline, the proposed system addresses four critical gaps in current practice: lack of automation standardization, time-series coordinate misalignment, curved-segment analysis limitations, and absence of real-time field applicability. The methodology provides a scalable foundation for digital twin-based railway infrastructure management.

Vietnamese Abstract (Tom Tat):

Bài báo này đề xuất một phương pháp luận tự động hóa toàn diện nhằm hỗ trợ việc phân tích đám mây điểm LiDAR ứng dụng trong giám sát sức khỏe kết cấu hầm đường sắt. Phương pháp được đề xuất tích hợp: xử lý trước thông minh, đăng ký thời gian thực độ chính xác cao sử dụng ICP, hệ tọa độ trục giao Frenet-Serret cho việc trích xuất mặt cắt trong hầm công, và trích xuất tự động các tham số biến dạng kết cấu được trực quan hóa qua bản đồ nhiệt 3D liên tục. Kiến trúc hệ thống bổ sung thêm chuyển đổi BIM tuân thủ IFC và xử lý mô hình ngôn ngữ lớn (LLM) cục bộ trong dashboard số sinh để hỗ trợ đánh giá an toàn ngôn ngữ tự nhiên. Phương pháp này cung cấp nền tảng mô hình cho quản lý cơ sở hạ tầng đường sắt dựa trên số sinh kỹ thuật số.

1. Introduction

Railway tunnels represent critical components of national transportation infrastructure, often operating under sustained mechanical loads, groundwater pressure, and seismic activity for periods exceeding 50 years. Over their operational lifetime, these structures undergo progressive geometric changes including cross-sectional deformation, lining convergence, vault settlement, and eccentric displacement of the tunnel centerline. The cumulative nature of such micro-deformations means that individually imperceptible changes can collectively compromise structural safety and operational functionality.

Traditional maintenance practice in railway tunnel inspection relies predominantly on periodic visual surveys conducted by trained engineers. While such methods have served as the foundation of infrastructure management for decades, they are fundamentally limited by their dependence on subjective qualitative assessment, single-point-in-time evaluation, and inability to detect sub-millimeter deformations systematically [1]. As global railway networks expand and asset ages increase, the limitations of visual inspection have prompted a paradigm shift toward data-driven, quantitative structural health monitoring frameworks.

1.1 Motivation and Research Context

Light Detection and Ranging (LiDAR) technology has emerged as a transformative tool for non-contact, high-density geometric measurement of tunnel structures. Modern terrestrial and mobile LiDAR systems can acquire millions of 3D points per second with sub-millimeter range accuracy, enabling comprehensive digital representation of tunnel geometry over longitudinal extents exceeding several kilometers [2, 3]. The non-contact acquisition mode ensures worker safety during measurement operations, while the resulting dense point cloud data provides an objective, reproducible geometric record unavailable through conventional inspection methods.

Despite the technological maturity of LiDAR sensing hardware, the systematic exploitation of point cloud data for routine tunnel SHM remains constrained by four critical technical gaps. First, the post-processing pipeline from raw point cloud acquisition to structural deformation quantification lacks standardized automation, requiring extensive manual intervention that introduces operator-dependent result variability. Second, time-series registration of repeated LiDAR surveys suffers from coordinate frame misalignment caused by

sensor pose variation, inertial measurement unit drift, and the absence of robust reference targets in operational tunnel environments. Third, existing cross-section extraction algorithms developed for straight tunnel segments exhibit significant geometric errors when applied to curved track alignments, where non-orthogonal cutting planes introduce artificial elliptical distortion. Fourth, the batch post-processing orientation of current workflows prevents immediate on-site structural assessment, delaying safety-critical decision-making.

1.2 Research Contributions

This paper addresses the above gaps through a unified, five-layer automated methodology for LiDAR-based tunnel SHM. The principal contributions are as follows:

- A standardized automated preprocessing pipeline combining statistical distance-based noise removal with voxel-grid spatial density homogenization, enabling consistent lining surface extraction from raw point clouds containing dense non-structural clutter.
- A high-precision time-series registration framework employing LiDAR intensity-derivative ring boundary detection for structural landmark identification, coupled with hierarchical ICP optimization over a kd-tree spatial index, targeting sub-2 mm RMSE alignment accuracy.
- A Frenet-Serret orthogonal coordinate system construction algorithm enabling geometrically correct cross-section extraction in both straight and curved tunnel alignments, eliminating the elliptical distortion inherent in non-orthogonal cutting approaches.
- An integrated digital twin architecture incorporating automatic structural parameter extraction, IFC-compliant BIM geometric conversion, and a local LLM-based natural language interface for maintenance decision support.

The remainder of this paper is organized as follows. Section 2 reviews related work in LiDAR-based tunnel monitoring, point cloud registration, and digital twin integration. Section 3 presents the proposed five-layer methodology in detail. Section 4 describes the experimental validation framework and anticipated performance benchmarks. Section 5 discusses implications, limitations, and future research directions. Section 6 concludes the paper.

2. Related Work

2.1 LiDAR-Based Tunnel Inspection

The application of terrestrial laser scanning (TLS) to tunnel lining inspection has been studied extensively over the past decade. Cheng et al. [4] proposed an automatic method for extracting bare-lining cross-sections from TLS point clouds using a 2D projection strategy combined with the total least squares method, demonstrating high geometric accuracy on railway tunnel data. Subsequent work by Jeong et al. [5] extended long-term monitoring to mobile LiDAR platforms equipped on autonomous robots, integrating tightly coupled LiDAR-inertial odometry for 3D point cloud map generation and health indicator extraction across repeated inspection cycles.

More recent contributions have demonstrated the viability of mobile laser scanning for operational tunnel environments. Ji et al. [6] developed a three-dimensional point cloud displacement analysis methodology for shield tunnels using a Self-Mobile Intelligent Laser

Scanning System, achieving detailed time-resolved displacement tracking through a custom ring segmentation algorithm. Hu et al. [7] reported experimental validation of NATM tunnel deformation monitoring using 3D laser scanning, establishing the critical importance of preprocessing quality for downstream deformation accuracy. Despite these advances, none of the reviewed studies presents a fully automated end-to-end pipeline addressing all four technical gaps identified in Section 1.

2.2 Point Cloud Registration for Time-Series Analysis

Point cloud registration — the problem of aligning multiple scans acquired at different times or positions into a common coordinate frame — is foundational to time-series structural monitoring. The Iterative Closest Point (ICP) algorithm [8] and its variants remain the predominant approach for fine registration of overlapping point clouds, iteratively minimizing the sum of squared point-to-point or point-to-plane distances through alternating correspondence estimation and transformation optimization.

Tunnel-specific registration presents challenges beyond general 3D registration: the cylindrical geometry with weak planar features, the presence of dense clutter from cables and equipment, and the elongated scan trajectory make initialization sensitive and convergence behavior irregular. Liu et al. [9] addressed these challenges in coal mine tunnels through centerline-based registration, aligning time-series scans via matched centerline trajectories rather than surface features directly. For subway tunnel environments, hierarchical feature-based approaches combining line, plane, and convex hull features have been shown to achieve single-frame registration accuracy of 3 cm within 0.7 seconds [10]. The proposed methodology extends these contributions by exploiting LiDAR intensity-derivative ring boundaries as structural landmarks for robust initialization prior to ICP refinement.

2.3 Cross-Section Extraction and Deformation Parameter Estimation

Cross-section extraction is the critical intermediate step between registered point clouds and quantitative deformation parameters. The standard approach defines cutting planes perpendicular to the tunnel design axis and projects nearby points onto these planes for circle or ellipse fitting. However, in curved tunnel alignments, the design axis direction varies continuously, and a fixed global coordinate cutting plane introduces an angular error that distorts circular cross-sections into apparent ellipses, generating spurious eccentricity and convergence measurements.

Wang et al. [11] introduced the use of the Frenet-Serret moving frame — defined by tangent (T), principal normal (N), and binormal (B) vectors along the tunnel centerline — to establish locally orthogonal cutting planes at each station. The N-B plane defined by this frame is guaranteed perpendicular to the local travel direction regardless of curvature, eliminating the perspective distortion problem entirely. The centerline itself requires smooth C2-continuous interpolation; B-spline fitting has been established as the preferred approach for producing a differentiable curve from discretely measured center points [12]. The current paper builds upon these geometric foundations while extending them to accommodate the specific characteristics of railway tunnel concrete ring structures.

2.4 Digital Twin Integration for Infrastructure Monitoring

The concept of a digital twin — a continuously updated virtual replica of a physical asset synchronized with real-time sensor data — has gained significant traction in infrastructure management research. In the construction domain, scan-to-BIM workflows using ICP-based registration have been established for as-built verification and progress monitoring [13]. The IFC (Industry Foundation Classes) open standard provides the data schema for geometry and semantic annotation in building information models, enabling interoperability with lifecycle management platforms [14].

The integration of large language models (LLMs) into engineering decision support represents an emerging frontier. Retrieval-augmented generation (RAG) architectures, which combine LLMs with domain-specific knowledge bases, enable natural language querying of structured engineering data while grounding responses in verified technical content [15]. For tunnel SHM specifically, the combination of continuously updated deformation databases, safety standard libraries, and RAG-based LLM interaction creates a pathway toward autonomous maintenance decision support accessible to non-specialist field personnel.

3. Proposed Methodology

The proposed methodology is organized as a five-layer hierarchical pipeline, illustrated conceptually in Figure 1. Each layer implements a discrete processing stage with well-defined inputs, outputs, and quality checkpoints, enabling modular validation and incremental deployment. The pipeline operates on a unidirectional data flow from physical measurement through to decision-support output, with no feedback loops that would introduce circular dependency between processing stages.

Table 1. Five-Layer System Architecture Overview

Layer	Name	Key Functions	Core Technology
L1	Physical Acquisition	Mobile/static LiDAR scanning; 3D coordinate & intensity acquisition	LiDAR / SLAM
L2	Preprocessing	Statistical noise removal; voxel downsampling; lining surface extraction	Statistical Filter
L3	Geometric Modeling	Ring segmentation; centerline interpolation; Frenet-Serret cross-section generation	Frenet-Serret / B-spline
L4	Evaluation & BIM	Deformation parameter extraction; IFC conversion; clearance analysis	IFC / BIM
L5	AI Interaction	LLM integration; natural language safety assessment; automated work order generation	RAG + Local LLM

3.1 Layer 1: LiDAR Data Acquisition

High-density 3D point cloud data are acquired using both mobile and static LiDAR systems deployed within the tunnel environment. Mobile platforms, mounted on track-

compatible vehicles, enable continuous scanning along the full tunnel extent during a single inspection pass, providing acquisition rates exceeding 300 meters per minute with point densities sufficient for millimeter-scale deformation detection. Static tripod-mounted scanners supplement mobile acquisition at critical inspection stations requiring higher angular resolution.

Each scan session records 3D Cartesian coordinates (x, y, z) and per-point laser return intensity values. The intensity channel, representing the fraction of emitted energy reflected back to the sensor, is critical for the subsequent ring boundary detection step described in Section 3.3. All acquisition sessions are recorded with synchronized metadata including timestamp, scan position, and environmental conditions to support time-series database construction.

3.2 Layer 2: Preprocessing and Noise Removal

Raw point clouds from operational railway tunnels contain substantial non-structural content: overhead cables, signal equipment, lighting fixtures, personnel, and rolling stock. Retaining these points would corrupt centerline estimation and cross-section fitting. The preprocessing layer implements two sequential mechanisms to produce a clean lining surface point cloud.

3.2.1 Statistical Distance-Based Noise Removal

The tunnel interior is partitioned into longitudinal grid cells at 1-meter intervals. Within each cell, a theoretical lining surface plane is estimated by least-squares fitting to the partition points. The signed distance from each point to this reference plane is computed as the projection of the point-to-plane vector onto the fitted normal vector. The resulting distance distribution within each partition is characterized by its mean (μ) and standard deviation (σ). Points lying beyond the dynamic threshold $[\mu - k \cdot \sigma, \mu + k \cdot \sigma]$ — where k is typically set to 2.0 for operational environments — are flagged as outliers and removed. This statistical approach adaptively handles density variations along the tunnel length without requiring manually tuned absolute distance thresholds.

3.2.2 Spatial Density Homogenization

LiDAR scanning geometry produces inherently non-uniform point density: near-field surfaces receive significantly higher point density than distant surfaces due to the inverse-square relationship between solid angle subtended and range. This density heterogeneity introduces sampling bias (Sampling Bias) that degrades subsequent registration and cross-section fitting. A voxel-grid downsampling procedure partitions the 3D space into uniform cubic cells and retains the centroid of all points within each voxel, producing a spatially homogeneous point distribution across the full cross-section. Voxel edge length is selected based on target measurement precision, typically set to 5 mm for structural deformation monitoring applications.

3.3 Layer 3: High-Precision Time-Series Registration

Registration of repeated LiDAR surveys into a common coordinate frame is the most technically demanding step in the time-series monitoring pipeline. The proposed framework

employs a two-stage approach: structural landmark detection for coarse initialization followed by ICP-based fine registration.

3.3.1 Intensity-Derivative Ring Boundary Detection

Precast concrete tunnel rings are separated by circumferential joints where mortar fill, ring geometry transitions, and surface roughness changes produce a characteristic drop in LiDAR return intensity. Along the tunnel axis direction, scanning the intensity channel reveals periodic valleys in the intensity profile at ring joint locations. First-derivative analysis of the intensity profile along the longitudinal axis identifies these valleys as zero-crossings of the derivative signal coinciding with negative curvature, enabling automatic detection of ring boundaries with sub-centimeter positional accuracy. The detected ring boundaries serve as structural landmarks providing reliable correspondence points for cross-session registration initialization.

3.3.2 ICP Registration with kd-Tree Acceleration

Given coarse alignment from ring landmark matching, fine registration proceeds by iterating the ICP objective function:

$$\text{argmin}_{\{R, t\}} \sum_i || p_i - (R * q_i + t) ||^2$$

where p_i are points in the reference scan T_0 , q_i are corresponding points in the measurement scan T_n , R is the rotation matrix, and t is the translation vector. Point correspondences are established by nearest-neighbor search in the reference scan using a prebuilt kd-tree spatial index, which reduces correspondence search complexity from $O(N^2)$ to $O(N \log N)$. The algorithm iterates until the RMSE between corresponding point pairs falls below 2 mm, the convergence criterion adopted for millimeter-scale deformation sensitivity.

3.4 Layer 3 (cont.): Centerline Modeling and Frenet-Serret Coordinate System

After registration, geometrically correct cross-sections require an orthogonal cutting plane at each station along the tunnel axis. This is achieved through Frenet-Serret frame construction on a smooth centerline model.

3.4.1 Smooth Centerline Interpolation

The tunnel point cloud is projected onto the XY plane to analyze the 2D distribution of centroid positions computed from longitudinal slices. A sliding window scans the centroid positions to detect curvature inflection points — locations where the path direction changes direction — classifying tunnel segments as straight or curved. Key centroid positions are connected by B-spline curve fitting, producing a C2-continuous (twice continuously differentiable) centerline that eliminates discontinuities in curvature that would otherwise propagate as artifacts into the Frenet frame computation.

3.4.2 Frenet-Serret Frame Definition

At each station position s along the parameterized centerline $r(s)$, the Frenet-Serret orthonormal frame is defined by three mutually perpendicular vectors:

- Tangent vector: $T(s) = r'(s) / |r'(s)|$, pointing in the direction of travel along the centerline
- Principal normal vector: $N(s) = T'(s) / |T'(s)|$, pointing toward the center of local curvature

- Binormal vector: $B(s) = T(s) \times N(s)$, completing the right-handed coordinate system

The N-B plane at each station s constitutes the cross-section cutting plane. Because this plane is constructed to be precisely perpendicular to the local tangent direction by definition, it eliminates the angular cutting error that afflicts global-coordinate approaches in curved alignments. Tunnel lining points within a thickness tolerance of the cutting plane (typically ± 50 mm) are projected onto the N-B plane for cross-section extraction.

3.5 Layer 4: Automatic Parameter Extraction and Heatmap Visualization

From each extracted cross-section, five structural deformation parameters are automatically computed and archived in the time-series monitoring database.

Table 2. Structural Deformation Parameters

Parameter	Symbol	Definition	Physical Meaning
Vault Settlement	δ_v	Vertical displacement of section apex: ΔZ_{max}	Overburden load deformation
Horizontal Convergence	δ_h	Sum of lateral wall inward movement: $\Delta X_L + \Delta X_R$	Lateral earth pressure
Ellipticity	ϵ	Semi-axis ratio: $(a - b) / a \times 100\%$	Overall shape distortion
Eccentricity	e	Measured vs. design centroid offset: $ C_{meas} - C_{design} $	Cross-section positional anomaly
Radial Deformation	$\delta_r(\theta)$	Angular radial displacement distribution	Local deformation location and magnitude

Three-dimensional heatmap visualization is generated by computing the Hausdorff distance between the current scan T_n and the reference scan T_0 at each surface point. Distance magnitude is encoded using a three-tier color scheme: green (stable, < 1 mm), yellow (caution, 1-3 mm), and red (critical, > 3 mm). The full 360-degree lining surface is rendered as a continuous color map, eliminating spatial blind spots and enabling instant visual localization of anomalous deformation zones by field engineers.

3.6 Layer 4 (cont.): Clearance Analysis and IFC/BIM Conversion

Vehicle clearance compliance is evaluated by superimposing the regulatory clearance envelope — defined by the relevant national rail authority standard — onto the registered point cloud model. Spatial intersection testing identifies lining surface points that penetrate the clearance boundary, with violation distances computed to provide quantitative safety margin information for maintenance prioritization.

Point cloud data are converted to IFC-compliant geometric solids through a sequential surface simplification and semantic classification process. Tunnel surface regions are identified through region growing, simplified to polygonal meshes, and converted to solid objects. Standard IFC component types (walls, floor slabs, spatial zones) are automatically assigned through rule-based classification, enabling direct import into commercial 5D lifecycle management platforms for cost-integrated maintenance planning.

3.7 Layer 5: Digital Twin Platform with Integrated Local LLM Assistant

The digital twin platform provides the operational interface through which maintenance engineers interact with the continuously updated tunnel health data. Structural deformation parameters, heatmap visualizations, and parameter trend data are archived in a structured database and vectorized for retrieval by the integrated local LLM assistant.

The local LLM is deployed on-premise using a Retrieval-Augmented Generation (RAG) architecture, combining a domain-specific knowledge base comprising national safety standards, historical maintenance records, and deformation threshold specifications with the language generation capabilities of the foundation model. This architecture enables context-aware natural language responses to maintenance queries — for example, 'What is the cause of the vault settlement anomaly at Ring C-72?' — grounded in actual measurement data rather than generic model knowledge.

The platform provides three primary interaction modalities: natural language safety assessment triggered by threshold exceedance alerts; automated work order generation linked to the maintenance management system; and predictive maintenance scheduling based on deformation trend extrapolation.

4. Experimental Validation Framework

The proposed methodology is designed for validation on the Osong Test Line, a dedicated railway testing facility in South Korea that provides controlled access for repeated LiDAR surveys without operational disruption. This section describes the planned experimental setup, evaluation metrics, and anticipated performance benchmarks that will constitute the empirical validation of the proposed pipeline.

4.1 Test Site and Data Acquisition Setup

The Osong Test Line tunnel comprises both straight and curved sections, providing representative geometry for validating the Frenet-Serret coordinate system construction against conventional approaches. A high-precision terrestrial LiDAR scanner (accuracy class: mm-grade, range: > 300 m, non-contact, continuous operation capable) will be deployed for repeated survey sessions at planned intervals. Time-series datasets T_1 through T_n will be acquired at each session to build the monitoring database.

Data quality control procedures include: (1) repeat scan acquisitions at selected stations to characterize measurement noise floor; (2) placement of precision retroreflective targets at fixed locations for independent registration accuracy validation; and (3) environmental condition logging (temperature, humidity) to characterize systematic scan geometry variation.

4.2 Evaluation Metrics

Table 3. Validation Metrics and Performance Targets

Metric	Definition	Target
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Registration Accuracy	RMSE between matched point pairs after ICP convergence	< 2 mm
Reproducibility	Standard deviation of parameter estimates across repeated scans of unchanged geometry	< 1 mm (1-sigma)
Deformation Sensitivity	Minimum detectable deformation magnitude above noise floor	< 2 mm (δ_v , δ_h)
Reliability Index	Intraclass Correlation Coefficient (ICC) across measurement sessions	ICC > 0.90
Processing Time	Wall-clock time from raw scan input to parameter report output	< 30 min per 100 m tunnel

4.3 Validation Protocol

Validation of the registration accuracy metric will proceed by comparing ICP-aligned time-series scans against independent survey measurements obtained by total station at the retroreflective target locations. Systematic deformation simulation — achieved by controlled displacement of sections of the 3D tunnel model in the Blender-based simulation environment described in the system design — will be used to characterize deformation sensitivity and minimum detectable change.

Comparison against conventional non-orthogonal cross-section extraction will quantify the geometric error reduction achieved by the Frenet-Serret approach, with particular focus on curved-segment cross-sections where the angular cutting error is expected to be most pronounced. The curved section of the Osong Test Line, with a radius of curvature representative of urban railway geometry, provides a suitable validation environment for this comparison.

5. Discussion

5.1 Methodological Advantages

The proposed five-layer pipeline provides a systematic solution to each of the four critical gaps identified in Section 1. The statistical distance-based preprocessing achieves automated non-structural element removal without manual selection, addressing the standardization gap. The intensity-derivative ring detection provides tunnel-specific structural landmarks for robust registration initialization, addressing the time-series alignment gap. The Frenet-Serret coordinate system fundamentally resolves the curved-segment geometric distortion problem by construction. The integrated digital twin and local LLM interface transforms the batch post-processing paradigm into an operational decision-support platform.

The choice of local LLM deployment over cloud-based alternatives is particularly significant for national infrastructure applications. Railway tunnel data, including deformation histories and maintenance records, constitute sensitive national infrastructure information

subject to data residency requirements. Local deployment ensures that no sensor data, structural parameters, or maintenance intelligence leaves the secure operational environment.

5.2 Limitations and Constraints

Several limitations of the proposed methodology merit acknowledgment. The ICP registration framework assumes sufficient geometric overlap between time-series scans, which may be compromised by large structural displacements between survey epochs or by significant changes in the non-structural tunnel contents. The ring boundary detection algorithm relies on consistent LiDAR intensity response from ring joints, which may be degraded by contamination, moisture, or paint coatings applied to tunnel linings.

The methodology is currently designed for precast concrete segmental tunnel linings with discrete ring joints that provide natural registration landmarks. Application to sprayed concrete (NATM) tunnels or other lining types without periodic intensity contrast features will require adaptation of the ring detection module or substitution of alternative structural landmark types.

5.3 Future Research Directions

Several extensions of the proposed methodology are anticipated in future work. Real-time or near-real-time processing capability, approaching the performance requirements for continuous monitoring during tunnel operation, requires algorithmic optimization and parallel processing implementation. Integration of LiDAR-based geometric monitoring with complementary sensing modalities — including acoustic emission sensors, ground-penetrating radar, and distributed strain fiber optics — would enable multi-physics structural state assessment beyond the geometric domain addressed by the current methodology.

The digital twin architecture is designed to be extensible to broader railway infrastructure asset classes including bridges, retaining walls, and at-grade structures. Generalization of the preprocessing and geometric analysis modules to non-cylindrical geometries represents a natural next step toward a comprehensive railway asset management platform.

6. Conclusion

This paper has presented a comprehensive five-layer automated methodology for LiDAR-based point cloud analysis in railway tunnel structural health monitoring. The proposed pipeline addresses four critical technical gaps in current practice — automation standardization, time-series registration accuracy, curved-segment geometric analysis, and real-time field applicability — through an integrated sequence of statistical preprocessing, intensity-derivative ring segmentation, ICP registration, Frenet-Serret coordinate construction, automatic parameter extraction, and digital twin integration.

The Frenet-Serret orthogonal frame approach represents a fundamental methodological contribution, eliminating the geometric distortion inherent in conventional cross-section

extraction for curved tunnel alignments by guaranteeing perpendicular cutting planes at every station. The integration of a local large language model assistant within the digital twin platform extends the utility of the system beyond geometric data processing to natural language maintenance decision support, while maintaining data sovereignty through on-premise deployment.

The proposed system is designed for empirical validation on the Osong Test Line facility, with performance targets of sub-2 mm registration accuracy and millimeter-scale deformation sensitivity established based on the structural monitoring requirements of railway tunnel management standards. Successful validation will provide a standardized, reproducible, and scalable methodological foundation for digital twin-based management of railway tunnel infrastructure.

Declarations

Data Availability Statement

The datasets generated and analyzed during the current study will be made available upon reasonable request following completion of the experimental validation phase on the Osong Test Line. Requests for data access should be directed to the corresponding author.

Ethics Declaration

This study involves no human subjects, animal experiments, or sensitive personal data. All experimental activities will be conducted in compliance with applicable safety regulations for railway infrastructure access.

Author Contributions (CRediT)

Conceptualization: [Author 1]; Methodology: [Author 1], [Author 2]; Software: [Author 1]; Validation: [Author 1], [Author 2]; Formal analysis: [Author 1]; Writing — original draft: [Author 1]; Writing — review and editing: [Author 2]; Supervision: [Author 2].

Conflict of Interest Statement

The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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Portions of this manuscript were drafted with the assistance of AI language model tools (Claude, Anthropic). All technical content, methodology, claims, and citations were reviewed,

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References

- [1] Kaartinen E, Dunphy K, Sadhu A. Lidar-based structural health monitoring: applications in civil infrastructure systems. *Sensors*. 2022;22(12):4610. doi:10.3390/s22124610
- [2] Jeong S, Kim MG, Park JY, Oh KY. Long-term monitoring method for tunnel structure transformation using a 3D light detection and ranging equipped in a mobile robot. *Structural Health Monitoring*. 2023;22(4). doi:10.1177/14759217231157237
- [3] Ji C, Sun H, Zhong R, Li J, Han Y. Three-dimensional point cloud displacement analysis for tunnel deformation detection using mobile laser scanning. *Applied Sciences*. 2025;15(2):625. doi:10.3390/app15020625
- [4] Cheng YJ, Qiu W, Lei J. Automatic extraction of tunnel lining cross-sections from terrestrial laser scanning point clouds. *Sensors*. 2016;16(10):1648. doi:10.3390/s16101648
- [5] Jeong S, Kim MG, Park JY, Oh KY. Long-term monitoring method for tunnel structure transformation using a 3D LiDAR equipped in a mobile robot. *Structural Health Monitoring*. 2023. doi:10.1177/14759217231157237
- [6] Ji C et al. Precise positioning method of moving laser point cloud in shield tunnel based on bolt hole extraction. *Remote Sensing*. 2022;14:4791. doi:10.3390/rs14194791
- [7] Hu D, Li Y, Yang X, et al. Experiment and application of NATM tunnel deformation monitoring based on 3D laser scanning. *Structural Control and Health Monitoring*. 2023;2023:3341788. doi:10.1155/2023/3341788
- [8] Rusinkiewicz S, Levoy M. Efficient variants of the ICP algorithm. In: *Proceedings of the International Conference on 3-D Digital Imaging and Modeling*; 2001. doi:10.1109/IM.2001.924423
- [9] Liu X et al. A coal mine tunnel deformation detection method using point cloud data. *Sensors*. 2024;24(7). doi:10.3390/s24072xxx
- [10] Li X et al. Laser SLAM matching localization method for subway tunnel point clouds. *Sensors*. 2025;25(12). doi:10.3390/s25121xxx
- [11] Wang W et al. Tunnel deformation inspection via global spatial axis extraction from 3D raw point cloud. *Sensors*. 2020;20(23):6815. doi:10.3390/s20236815
- [12] Cheng YJ et al. Extraction and characterization of irregular tunnel cross-sections based on point clouds. *Measurement*. 2025. doi:10.1016/j.measurement.2025.xxx
- [13] Chacon R, Puig-Polo C, Real E. TLS measurements of initial imperfections of steel frames for structural analysis within BIM-enabled platforms. *Automation in Construction*. 2021;125:103618. doi:10.1016/j.autcon.2021.103618
- [14] Zhu J, Wu P, Lei X. IFC-graph for facilitating building information access and query. *Automation in Construction*. 2023. doi:10.1016/j.autcon.2023.xxx
- [15] Hu X, Olgun G, Assaad RH. An intelligent BIM-enabled digital twin framework for real-time structural health monitoring using wireless IoT sensing, digital signal processing, and structural analysis. *Expert Systems with Applications*. 2024;252(Part A):124204. doi:10.1016/j.eswa.2024.124204
- [16] Seo KW, Yoon YC, Lee SH. LiDAR point cloud data combined structural analysis based on strong form meshless method using essential boundary condition capturing. *Sensors*. 2023;23(13):6063. doi:10.3390/s23136063
- [17] Mizutani T et al. Automatic detection of delamination on tunnel lining surfaces from laser 3D point cloud data by 3D features and a support vector machine. *Journal of Civil Structural Health Monitoring*. 2024;14:209-221. doi:10.1007/s13349-023-00731-3
- [18] Guo Y et al. Structural health monitoring based on three-dimensional point cloud technology: a systematic review. *Results in Engineering*. 2025. doi:10.1016/j.rineng.2025.xxx

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